Algorithm 1 Failure Probability Estimation via CEM

Input: Failure indicator f, original distribution P, sample size N, smoothing coefficient ζ, iterations T, confidence level α, defensive sampling coefficient λ
Output: Failure probability estimate $\hat{\mu}$, confidence interval $[\hat{\mu}_{\text{low}}, \hat{\mu}_{\text{high}}]$
// CEM for updating Q
Initialize $Q_0(Z_i)$ as uniform over $\mathcal{Z}_i$ for all $i = 1, \ldots, d$
for $t = 1$ to T **do**
  // Sampling
  Sample $Z_1, \ldots, Z_N \sim Q_t(Z) = \prod_{i=1}^{d} Q_t(Z_i)$
  // Elite set selection
  Define elite set $\mathcal{E}_t \leftarrow \{n : f(Z_n) = 1\}$
  if $\mathcal{E}_t = \emptyset$ **then**
    Set $Q_{t+1} \leftarrow Q_t$ and **continue**
  end if
  // Weighted update
  for each dimension i and value $j \in \mathcal{Z}_i$ **do**
    Compute importance weights $w(Z_n) \leftarrow P(Z_n)/Q_t(Z_n)$
    $Q'_{t+1}(Z_i)_j \leftarrow \frac{\sum_{n \in \mathcal{E}_t} w(Z_n)\, \mathbb{1}\{z_{n,i}=j\}}{\sum_{n \in \mathcal{E}_t} w(Z_n)}$
  end for
  // Smoothing
  $Q_{t+1} \leftarrow \zeta Q'_{t+1} + (1-\zeta) Q_t$
end for
Apply defensive sampling $\tilde{Q}_T(Z) \leftarrow (1-\lambda) Q_T(Z) + \lambda P(Z)$
// Estimation with confidence bounds
$\hat{\mu}, [\hat{\mu}_{\text{low}}, \hat{\mu}_{\text{high}}] \leftarrow \textsc{ComputeEstimate}(f, P, \tilde{Q}_T, N, \alpha)$
Return $\hat{\mu}$, $[\hat{\mu}_{\text{low}}, \hat{\mu}_{\text{high}}]$

Failure probability estimation via Q learned from CEM. To begin with, we model Q as a fully factorized distribution:

$$Q(Z) = \prod_{i=1}^{d} Q(Z_i), \tag{4}$$

where each $Q(Z_i)$ is a categorical distribution over $\mathcal{Z}_i$. This independence assumption provides a tractable approximation to model the joint parameter space. We initialize $Q_0(Z)$ as a uniform distribution over each parameter dimension $Z_i \in \mathcal{Z}_i$.

We then apply the CEM followed by importance sampling for estimating true failure probability with Q, as described in Algorithms 1 and 2.

6 Order-of-Magnitude Inference Efficiency Gains

Setup for CEM For CEM, we use fixed hyperparameters across all models and templates: T=10 iterations, smoothing coefficient ζ=0.1, and defensive sampling coefficient $\lambda \in \{0.1, 0.3, 0.5\}$. The CEM sample size N, the number of data for updating Q at each CEM iteration t, is set as a hyperparameter, ranging between 0.5K and 100K.

Setup for estimation. For the final estimation, we vary the evaluation sample size M between 1K and 1M (million). We define the *number of inferences* as the number of samples from P or Q (one inference per Z, independent of K). Because the $T \times N$ samples drawn during CEM can be reused directly as importance sampling estimation samples, the total number of inferences under Q is $\max(T \times N,\ M)$: if the CEM draws already reach or exceed M, no additional inferences are needed; otherwise, $M - T \times N$ samples are drawn from $\tilde{Q}_T$. For P, the cost is simply M inferences with no CEM phase.

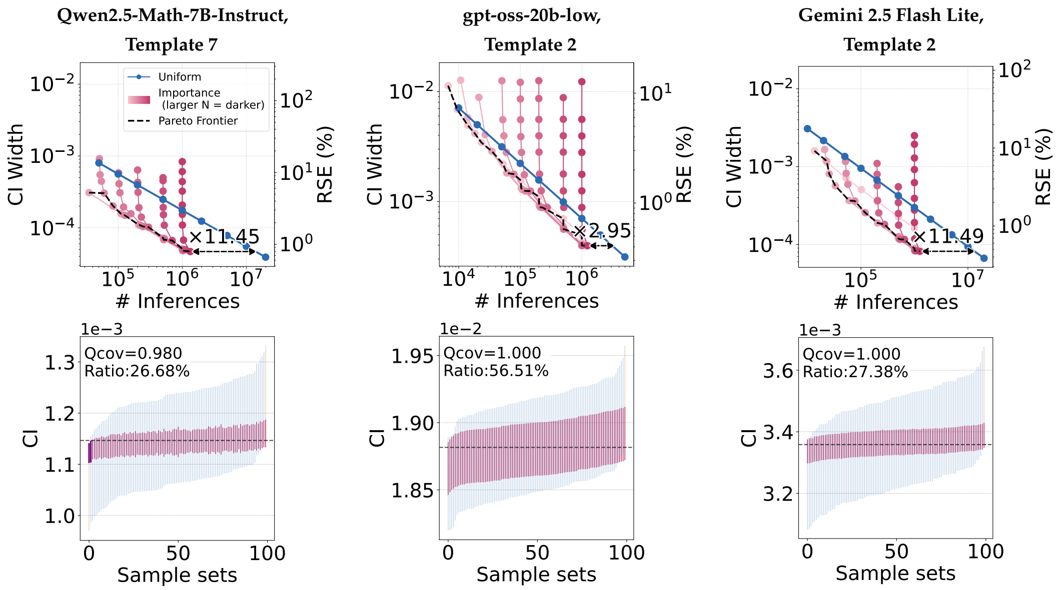

Figure 4: **Upper row: CI width vs. number of inferences.** Each solid pink lines connects estimations with varying number of inferences (pink dots) under importance sampling with same number of overhead inferences for CEM (N), and darker pink means larger N. Similarly, solid blue line for uniform sampling. The Pareto-frontier (dotted line) is the lower-left envelope across all $(N, \#\,\text{evaluation})$ combinations, which achieves a tight CI width, i.e., relative standard error (RSE) of around 1%, at a much smaller inference budget compared to uniform sampling **Lower row: sorted CIs over** 100 **sample sets.** At the evaluation size of the bottom-right Pareto-frontier dot, Q (pink) yields much narrower confidence bounds than P (blue) for the same sample size, while maintaining coverage close to 99%. We highlight the intervals that fail to cover the population mean in dark pink.

Setup for evaluation. For all evaluations, we construct 99% confidence intervals (significance level $\alpha = 0.01$) using the normal approximation applied to the importance-weighted estimator $\hat{\mu}_Q$. To verify the unbiasedness of our estimator, we draw 100 independent sample sets for each sample size and report the estimates whose coverage falls within $99 \pm 1\%$ (i.e., 1% tolerance of the nominal coverage level). For calculating coverage, we use $\hat{\mu}_P$ computed using 1B bootstrap samples from P, which converges to the unknown true mean.

We compare against uniform sampling from P with confidence bounds computed via the Clopper-Pearson binomial exact method (Clopper & Pearson, 1934). To quantify the inference efficiency obtained by importance sampling, we present the *sampling efficiency gain*, calculated by the number of inferences under P divided by that under Q (i.e., $\max(T \times N, M)$) to obtain the same CI width. Across N and evaluation sizes, we report the gains of the Pareto-frontier in the CI width vs. number of inferences trade-off (i.e., dotted line in Figures 1 and 4). We additionally report *RSE* defined as $\text{RSE} = \frac{\text{SE}}{\hat{\mu}}$ to evaluate the tightness of the confidence bounds.

Failure-prone sampling saves an order of magnitude in inferences. Our results demonstrate that the learned distribution Q, which concentrates on frequent failure patterns, yields substantially tighter confidence bounds compared to uniform sampling under P. In Figure 1 and upper row of Figure 4, we present CI width under Q (pink) for varying evaluation sizes (dots connected by lines) with different N, the number of samples for CEM (different shades). The Pareto-frontier (dotted line), which connects the minimum number of inferences required to achieve each CI width, decreases much more rapidly under Q as the number of inferences increases than under P (blue).

This indicates that, for a tight confidence bound such as an RSE of 1%, importance sampling under Q requires significantly fewer inferences than uniform sampling from P. Specifically,

	$\lambda = 0.1$						$\lambda = 0.3$						$\lambda = 0.5$					
	Gains			**RSE** (%)			**Gains**			**RSE** (%)			**Gains**			**RSE** (%)		
Qwen2.5-Math-7B-Instruct																		
ID	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16
0	3.68	6.50	16.41	1.24	1.22	1.02	4.51	8.03	15.84	1.27	1.21	1.05	3.82	6.59	13.65	1.27	1.26	1.13
1	1.31	2.29	5.33	1.97	2.99	2.95	1.58	2.34	7.02	1.94	3.00	2.46	1.57	2.30	6.02	1.95	3.21	2.69
2	1.63	1.48	1.87	0.22	0.24	0.27	1.37	1.48	1.80	0.22	0.25	0.27	1.51	1.61	1.68	0.22	0.26	0.29
3	1.07	1.46	1.75	0.43	0.63	0.80	1.25	1.50	1.49	0.43	0.63	0.81	1.24	1.25	1.64	0.43	0.64	0.83
4	1.34	1.38	1.22	0.31	0.36	0.40	1.30	1.38	1.23	0.31	0.36	0.40	1.29	1.33	1.41	0.31	0.37	0.41
5	2.15	3.67	6.93	1.20	1.77	5.05	2.51	4.28	5.40	1.14	1.87	2.81	2.32	3.69	5.62	1.19	1.85	2.78
6	5.36	15.84	17.12	0.44	0.44	0.43	6.40	12.87	15.92	0.49	0.49	0.47	5.80	9.46	11.74	0.56	0.56	0.54
7	3.22	8.02	11.45	1.08	0.95	0.79	3.53	8.45	14.66	1.03	0.89	0.80	4.44	6.42	11.59	1.03	0.96	0.87
8	1.91	—	—	4.97	—	—	2.73	—	—	4.09	—	—	2.45	—	—	4.57	—	—
gpt-oss-20b-low																		
ID	K=8	K=16	K=24	K=8	K=16	K=24	K=8	K=16	K=24	K=8	K=16	K=24	K=8	K=16	K=24	K=8	K=16	K=24
0	3.34	3.81	3.63	2.56	2.95	3.92	2.95	3.79	6.77	2.52	3.10	2.85	2.71	4.45	3.54	2.63	3.11	4.15
2	2.27	2.70	2.95	0.28	0.37	0.41	2.28	2.86	3.28	0.30	0.39	0.43	1.93	2.35	2.81	0.32	0.42	0.46
4	1.30	1.93	2.65	0.79	1.19	1.36	1.55	1.69	2.30	0.79	1.16	1.34	1.53	1.91	2.09	0.80	1.19	1.40
7	2.89	2.80	4.45	1.61	2.76	2.29	3.07	3.50	5.22	1.62	2.14	2.18	2.39	4.29	5.39	1.65	2.11	2.31
8	4.36	4.41	4.57	0.84	1.18	1.36	4.02	4.42	4.65	0.90	1.21	1.42	2.99	3.43	3.70	0.97	1.32	1.49
Gemini 2.5 Flash Lite																		
ID	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16	K=4	K=8	K=16
0	0.96	4.05	5.29	1.95	1.90	2.35	1.23	4.74	8.23	1.64	1.72	1.74	1.35	4.90	8.62	1.57	1.71	1.93
2	2.54	4.41	11.49	0.68	0.62	0.47	2.76	5.69	10.35	0.65	0.62	0.51	2.38	5.16	9.50	0.66	0.64	0.57
4	1.12	0.99	1.00	0.40	0.51	0.58	1.11	1.18	1.19	0.40	0.51	0.58	0.95	1.16	1.00	0.40	0.51	0.58
6	1.06	2.75	156.22	0.99	1.80	0.32	1.10	3.45	85.21	0.98	1.63	0.36	1.27	3.32	82.59	0.98	1.52	0.37

Table 2: Sampling efficiency gains and RSE (%) across templates, models, K, and λ for majority voting. Cell shading for gains is proportional to the gain value, i.e., the higher the darker, capped at gain ≥ 10. — denotes zero failures.

such inference efficiency gains span orders of magnitude across models and templates, e.g., 17.12× for Qwen2.5-Math-7B-Instruct on Template 6 ($K = 16$), 4.57× for gpt-oss-20b-low on Template 8 ($K = 24$), and 156.22× for Gemini 2.5 Flash Lite on Template 6 ($K = 16$).

These inference efficiency gains stem from the tight CIs obtained through importance sampling, as illustrated in Figure 4 bottom row. The plots compare the confidence bounds between P and Q across 100 independent sample sets (sorted by their center). Across evaluation sizes, we plot the bounds for the sample size corresponding to the bottom-right pink dot on the Pareto-frontier. Intervals under Q (pink) are much narrower than those under P (blue) for the same sample size. These 100 intervals faithfully bound the $\hat{\mu}_P$ (with ± 1 tolerance), confirming that our method meets the nominal 99% confidence level.

We report inference efficiency gains and their RSE across models and templates in Table 2. For each gain, we add darker shading for larger values for visibility. The results demonstrate that our method achieves consistent efficiency gains, where most exceed 2×. These estimations show substantially low RSE, where most are around just 1%, implying that we can estimate LLMs' true error probability with highly accurate estimation across templates and models.

We observe that inference efficiency gains improve as we use larger K for self-consistency decoding (Wang et al., 2023), i.e., shading becomes darker as K increases in gains. For instance, Qwen2.5-Math-7B-Instruct shows a gain of 3.68× with $K = 4$ increasing to 16.41× with $K = 16$ on Template 0, sharing the trends across setups. These results corroborate our empirical observations that systematic failures become more concentrated with higher K. In addition, overall gains and RSE remain consistent across different λ, which allows practitioners to use our method without careful hyperparameter tuning.

Similarly, efficiency gains are strongly correlated with failure rarity: as models become more saturated on a task, failures concentrate on a smaller subset of inputs, making importance sampling more effective. As a concrete example, Gemini 2.5 Flash Lite shows higher error rate ($\hat{\mu}_Q$) on Template 4, i.e., 2.59%, than others (0.335%-0.0328%) with $K = 16$ correspondingly, those high error-rate templates also exhibit lower TV distance (Table 1)